atomic< bool >
-goalActivitySeen_
-goalInFlight_
-isShutdownRequested_
SharedPtr
-end_
-initialized_
-rosInitialized_
processor_handle
-statusPub_
-processorHandle_
thread
-signalDetectorThread_
ExecutionModel
-executionModel_
vector< smacc2::ISmaccUpdatable * >
-updatableClients_
vector< std::vector < smacc2::ISmaccUpdatable * > >
-updatableStateElements_
atomic< int64_t >
-lastState_
SmaccFifoScheduler *
-scheduler_
double
-loop_rate_hz
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
SharedPtr
#nh_
shared_ptr< smacc2 ::introspection::SmaccState Info >
#currentStateInfo_
SmaccStatus
#status_msg_
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
#orthogonals_
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
#stateMachineInfo_
#eventsLogPub_
#stateMachinePub
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
recursive_mutex
-eventQueueMutex_
-m_mutex_
atomic< StateMachineInternal Action >
-stateMachineCurrentAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
-stateCallbackConnections
-smaccStateMachine_
-signalDetector_
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
-globalData_
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
-transitionLogHistory_
SMRunMode
-runMode_
int64_t
-stateSeqCounter_
vector< smacc2::ISmaccState * >
#currentState_
SharedPtr
#timer_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
vector< smacc2::introspection ::SmaccTransitionInfo >
+transitions_
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
+children_
type_info *
+tid_
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
+stateReactorsInfo
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
+eventGeneratorsInfo
int
+depth_
+stateIndex_
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
+staticBehaviorInfo
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
+stateMachine_
weak_ptr< smacc2::introspection ::SmaccStateInfo >
+parentState_
string
+demangledStateName
+fullStateName
#node_
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
vector< std::shared _ptr< smacc2::StateReactor > >
#stateReactors_
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
#eventGenerators_
shared_ptr< rclcpp ::Logger >
#logger_
#stateInfo_
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-currentState
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
-clientBehaviors_
vector< std::shared _ptr< smacc2::ISmaccClient > >
#clients_
-stateMachine_
-mutex_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
#components_
-stateMachine_
-orthogonal_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#currentPathName_
#odomFrame_
#odomTopicName_
#owner_
#stateMachine_
#m_mutex_
-poseFrameName_
-referenceFrame_
mutex
-m_mutex_
#robotPoseTimer_
#clearAngularDistanceThreshold
#clearPointDistanceThreshold
#recordAngularDistanceThreshold
#recordPointDistanceThreshold_
optional< rclcpp::Duration >
-periodDuration_
optional< rclcpp::Time >
-lastUpdate_
+frozenReferenceFrameTime
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const WrappedResult &), optional_last_value < typename boost::function _traits< void(const WrappedResult &)>::result_type >, int, std:: less< int >, function< void(const WrappedResult &)>, typename extended _signature< function_traits< void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
-actionMutex_
boost::signals2::signal < void(const Feedback &), optional_last_value < typename boost::function _traits< void(const Feedback &)>::result_type >, int, std ::less< int >, function< void (const Feedback &)>, typename extended _signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
-currentOrthogonal
#actionClientName_
< void()>
< void(const WrappedResult &)>
< void(const Feedback &)>
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
optional< std::future < int > >
-onEntryThread_
-onExitThread_
function< void()>
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
smacc2::SmaccSignal < void()>
-onFailure_
-onFinished_
-onSuccess_
optional< std::shared _future< typename GoalHandle ::SharedPtr > >
-lastRequest_
optional< std::shared _future< typename rclcpp _action::Client< ActionType >::CancelResponse::SharedPtr > >
-lastCancelResponse_
FeedbackCallback
-feedbackCallback_
optional< std::chrono ::milliseconds >
+serverTimeout
shared_ptr< ActionClient >
-client_
+onGoalAccepted_
+onGoalRejected_
+onActionAborted_
+onActionCancelled
+onActionSucceeded_
+postFeedbackEvent
+onActionFeedback_
smacc2::SmaccSignal < void(const WrappedResult &)>
+onNavigationAborted
+onNavigationCancelled
+onNavigationSucceeded_
function< void(const Feedback &)>
+postNavigationFeedbackEvent
smacc2::SmaccSignal < void(const Feedback &)>
+onNavigationFeedback_
function< void(const WrappedResult &)>
+postAbortedEvent
+postCancelledEvent
+postSuccessEvent
optional< std::string >
+actionServerName
+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent
CpActionClient< ActionType > *
-actionClient_
bool
+isInitialized
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
milliseconds
#goalResponseTimeout_
smacc2::client_core _components::CpActionClient < ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
cl_nav2z::components ::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
ResultCode
-resultConnectionsInitialized_
#actionClient_
#actionResult_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
shared_ptr< tf2_ros ::Buffer >
-tfBuffer_
shared_ptr< tf2_ros ::TransformListener >
-tfListener_
weak_ptr< tf2_ros:: Buffer >
-sharedTfBuffer_
weak_ptr< tf2_ros:: TransformListener >
-sharedTfListener_
mutex
-listenerMutex_
PoseStamped
-pose_
cl_nav2z::CpPose
+ CpPose()
+ CpPose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()
vector< nav_msgs::msg ::Path >
#pathStack_
vector< cl_nav2z::odom _tracker::CpOdomTracker ::PathInfo >
#pathInfos_
OdomTrackerStrategy
#strategy_
SharedPtr
#param_callback_handle_
#publishMessages
#robotPose_
SharedPtr
#odomSub_
#robotBasePathPub
#robotBasePathStacked Pub_
Path
#aggregatedStackPathMsg
#baseTrajectory_
WorkingMode
#workingMode_
optional< geometry _msgs::msg::PoseStamped >
#currentMotionGoal_
cl_nav2z::odom_tracker ::CpOdomTracker
+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
# onInitialize()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()
smacc2::client_behavior _bases::CbActionClientBehavior Base< TAction >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
< nav2_msgs::action ::NavigateToPose >
smacc2::client_behavior _bases::CbActionClientBehavior Base< nav2_msgs::action::NavigateToPose >
+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()
#navigationResult_
#nav2ActionInterface_
+goalChecker_
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# onNavigationActionSuccess()
# onNavigationActionAbort()
# onActionSuccess()
# onActionAbort()
optional< float >
+forwardSpeed
#forwardDistance_
optional< geometry _msgs::msg::Quaternion >
+forceInitialOrientation
cl_nav2z::CbNavigateForward Options
+options
#odomTracker_
#goalPose_
cl_nav2z::CbNavigateForward
+ CbNavigateForward()
+ CbNavigateForward()
+ CbNavigateForward()
+ ~CbNavigateForward()
+ onEntry()
+ onExit()
+ setForwardDistance()